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Clean Code Against Dirty Cash: A Survey on Anti-Money Laundering Techniques

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***Abstract*—Money laundering has become a significant challenge for financial institutions and regulatory authorities worldwide due to criminals who continue to exploit financial systems to hide illicit funds. Current anti-money laundering (AML) procedures still do not adequately combat current practices of money laundering with the help of advanced artificial intelligence (AI) and machine learning (ML) techniques. In addition, new methods of detecting anomalies in AML have benefited from recent developments in deep learning based on the use of transformer architectures, recurrent neural networks, and graph neural networks (GNN). The use of Generative Adversarial Networks (GAN) has improved the ability to train AML models through the use of synthetic transaction data created using the GANs. A major concern of AI-based AML systems is the "Black Box" nature of how these models make decisions, therefore raising questions of transparency and regulatory compliance. Furthermore, to mitigate this issue, researchers are looking to explainable methods like LIME (Local Interpretable Model Agnostic Explanations) and SHAP (Shapley Additive Explanations) as a means of improving the interpretability of models and ensuring compliance with financial regulations. In this review, we integrated the contributions of recent literature and findings from a critical survey of modern AI-driven AML systems to provide a comparative perspective on the pros and cons of AI-based AML systems in detecting cross-border financial anomalies. This review also provides a valuable resource for researchers, policymakers and financial analysts who wish to apply modern ML techniques to improve existing AML methods.**

***Index Terms*—Anti-money laundering, fraud detection in finance, machine learning, deep learning, anomaly detection, transformer models, graph neural networks (GNNs), and generative adversarial networks (GANs).**

I. Introduction

Money laundering is an important issue worldwide. It is a major facilitator of illegal activity such as corruption, financing of terrorism, and tax evasion. Therefore, the current methods of AML Systems to detect these complex frauds are ineffective because they rely on the establishment of rules and thresholds. As a result, these systems often have large numbers of false positives and are unable to identify new trends and techniques used to perpetrate money laundering [1] [2] [3]. In addition, manual detection by financial institutions is slowing the speed with which they can react to money laundering activities and identify and reflect new methods, techniques, and activities [4]. The greatest challenge faced by AML efforts is to detect anomalies associated with cross-border transactions where money laundering operatives use differences in the various regulatory requirements across the globe to conduct fraudulent activity [5]. Recent studies have highlighted the role of generating synthetic transaction data, improving model robustness, and reducing bias in training data through generative adversarial networks (GANs). Moreover, hybrid models that combine supervised and unsupervised learning techniques are being proposed to improve AML systems' adaptability to emerging money laundering techniques [6].

Disguising the illegitimate origins of unlawfully obtained cash to make them seem respectable is called the money laundering process. [7] [8]. This process includes three main stages or layers, as shown in Figure 1.

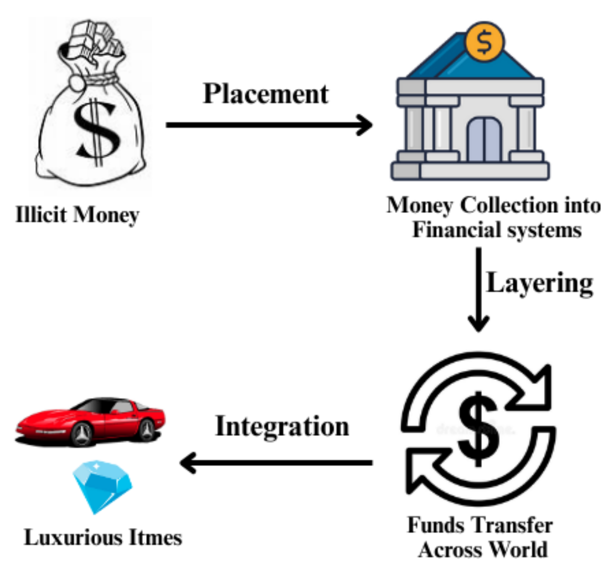

Fig. 1: Three Steps of Money Laundering.

The earliest phase of money laundering is getting acquired money into the legal financial system. Various techniques or approaches are followed to conceal the origins of laundered money, including placing cash in bank accounts, buying luxury items, and hiring payment services. To evade identification, criminals split a large quantity of money into smaller denominations, referred to as "smurfing." The second phase in money laundering comprises layering. It says that to prevent the original sources of illegal money from being identified, complex financial activities are designed, The penultimate phase of money laundering is integration. It entails reintegrating the money that was laundered into the legal economy in a way that makes it seem to have a valid source. In this, the money is employed to fund legal businesses, acquire new resources, and fund genuine organizations [9].

This paper reviews state-of-the-art ML and DL approaches to integrate these with AML by dedicating an effort to cross-border transaction monitoring, anomaly detection, and regulatory challenges. This study aims to provide insightful conclusions for researchers, policymakers, and financial institutions by analyzing recent advancements and their practical implications to improve AML approaches with AI-integrated solutions.

II. Overview of Machine Learning and Deep Learning TECHNIQUES in AML

Traditional rule-based learning depends on predefined rules and human-driven analysis, which have shown discrepancies in detecting efficient money laundering schemes. These approaches fail to detect emerging financial crime tactics and struggles and often generate high false positive rates [10]. Machine learning has secured remarkable improvement over traditional approaches in AML by plotting historical transaction data to identify illegal activities. For fraud detection models, ensemble learning techniques, support vector classifiers (SVMs), decision trees, and gradient boosting are used. But these models need labeled datasets, which are frequently limited due to privacy and regulatory caution [11]. To overcome this limitation, unsupervised learning algorithms, like anomaly identification methods and clustering algorithms, are being explored to detect hidden patterns in transaction data without being accustomed to labeled data. To identify anomalies with only a very small amount of data, here comes the role of semi-supervised learning. semi-supervised learning can work well in the case of data scarcity [12]. On the other hand, deep learning has shown excellent performance in detecting complex transaction patterns, especially in cross-border money laundering situations. Models such as Recurrent Neural Networks (RNNs) and Long Short-term Memory (LSTM) are used to monitor sequential data, capture temporal dependencies, and detect unusual behavior at a time [13]. Another progress in AML techniques is self-supervised learning, in which models learn meaningful transaction patterns without any need of explicit labels, which is very useful and efficient for anomaly detection, where labelled datasets are limited and highly sensitive.

III. Literature Review

There is a summarized differentiation of the key research publications presented in Table I, which highlights the methods and models used, outcomes, and key focus areas discussed in this section. Jensen and Losifidis [1] provided a statistical and machine learning-based review of AML, focusing on the strengths of supervised models in real-time fraud detection while cautioning on overfitting with imbalanced datasets.

Avacharmal and Pamulaparthyvenkata [2] specified various ML algorithms, including random forest, logistic regression, and XGBoost. Their comparative study showed ensemble models to have lower false positives and be able to adjust to the dynamic laundering strategies. Unsupervised clustering, especially K-means and DBSCAN, was utilized for discovering new patterns without the help of labeled data.

Bakry et al. [3] developed a hybrid SVM (Support Vector Machine) and decision tree model that combined both techniques into a single model to reduce the number of false alerts generated by AML systems.

Liu et al. [5] proposed a hybrid graph-based model to detect illegal and suspicious transactions based on inter-node relationships in a financial network using GNN (Graph Neural Network) technology to represent the transaction data in a graph.

Cheng et al. [7] developed a sequential model to monitor and classify money flow by identifying and flagging suspicious behaviours in transactions occurring over multiple accounts over extended periods of time.

Khan et al. [8] utilized LSTM (Long Short Term Memory) to identify anomalies occurring in cross-border transactions. The temporal dependencies in transaction sequence are captured by LSTM allowing them to detect newly developed laundering techniques as they emerge over time.

Zhang and Sun [9] developed transformer models, specifically BERT (Bidirectional Encoder Representations from Transformers) to monitor the textual data contained in financial documents, SAR, and customer profiles.

Afshar [13] developed the ITERADE model that is based on a transformer architecture to carry out anomaly detection using an ensemble approach. This new approach has demonstrated excellent performance at detecting instances of financial fraud and can be applied to case studies of Anti-money laundering (AML).

Lokanan and Sharma [15] contrasted the use of machine learning models and neural networks within banking systems focused upon AML. The study concentrated upon the function of Neural Networks (NNs), Convolutional Neural Networks (CNNs) and combinations of deep learning methodologies as a means to drive operational improvement in compliance and improve the efficiency of detection by lowering compliance operating expenses.

Bakhshinejad [20] further expanded this idea by using deep graph learning, particularly group-aware GNNs, to reveal community-based laundering strategies and circular fund movements.

TABLE I: Literature Review

AUTHOR	APPROACH	MODEL USED	FOCUS	RESULTS
Jensen et al. [1]	supervised Learning	Logistic regression, Random forest	Identifying suspicious financial transactions using historic data.	High accuracy in binary classification of fraudulent vs legitimate activities.
Avacharmal et al. [2]	Ensemble learning	XGboost, gradient boosting	Improving detection by combining weak learners.	Enhanced performance and reduced false positives.
Bakry et al. [3]	Unsupervised Learning	K-means clustering	Detecting unknown money laundering patterns	Discovering hidden patterns without labeled data.
Cheng et al. [5]	Graph-based learning	GNN (Graph neural network)	Detecting money laundering via transaction network analysis.	Effective in capturing complex transaction relationships.
Liu et al. [7]	Mobile AML framework	CNN, profiling	Real-time alerts	Scalable Hybrid system
Khan et al. [8]	Cross-border detection	CNN-GRU (Convolutional neural networks-gated concurrent units)	Sequential anomalies	High accuracy, lower false positives
Zhang et al. [9]	Unsupervised Deep learning	Autoencoders	Anomaly scoring	Improved anomaly detection
Afshar et al. [13]	deep learning	LSTM	Sequence analysis of transactions over time.	High performance in modelling temporal dependencies.
Lokanan et al. [15]	Transformer-based learning	BERT, financial transformer	Natural language processing of transactions and logs.	Improved interoperability and classification of textual AML data
Bakhshinejad et al. [20]	Hybrid approach	SVM+ Decision trees	Combining models to enhance generalization.	Achieved better precision and recall.

IV. Methodology

The flowchart presented in Figure 2 summarizes the sequence of steps taken to implement the AML detection model. The process followed these sequential steps: data collection and data preprocessing, feature engineering and data preparation, model training, and evaluation.

A. Data Collection

When detecting illegal acts like money laundering, transactional data is collected, evaluated, and disseminated. To collect the data as per Figure 2, financial institutions that are authorized submit records of their suspicious activity. Once the data is recorded, the payments are evaluated for patterns and anomalies, and the findings are then disseminated to enforcement authorities who are conducting ongoing investigations [14].

B. Preprocessing and Feature Engineering

Data preprocessing is to clean, etc., and encode categorical variables, to enable data ingestion into the model. Feature engineering involves the extraction, transformation and selection of the relevant features as predictors, for the model to provide greater accuracy [15] [16].

C. Model Selection

To detect anomalies in an AML system, a wide variety of ML and DL models are used. Each model has its own complexity, interpretability and ability to deal with financial data.

1) Support Vector Machines (SVM)

SVMs perform well on moderately high dimensional data and generally exhibit generalization capabilities to separate feature spaces with limited labeled data. Performance can be enhanced, for example, with techniques such as RUS or via hyperparameter tuning [17].

2) Random Forest

A collection of decision trees used for classification tasks. It automatically deals with missing data and helps reduce over fitting, so it is good for predicting financials .

3) Gradient Boosting

Sequentially fits a weak learner, minimizing errors. Good at tracing complex patterns financial patterns, decreasing false alarms.

4) Decision Trees

Translucent model structure helpful for compliance and classification. Gives a clear explanation to the investigator of how it came to the conclusion to flag a transaction [18].

5) Clustering Algorithms (e.g., KNN)

KNN identifies based on how close various data points are. It is computationally intensive on larger data while remaining interpretable in an AML context [19].

6) Recurrent Neural Networks (RNNs)

RNNs learn sequential dependencies. It could identify indicative patterns like layering and smurfing. It is more effective than other ML methods, but is less interpretable and requires higher computational resources.

7) Long Short-Term Memory (LSTM)

An RNN used for considering long term dependencies and design to learn from time-series and reduce false positives; therefore, it is a good tool for financial products [20] [21].

8) Graph Neural Networks (GNNs)

The best fit for modeling the relationships with transactional networks, especially to detect circularity and entity level behaviors and hidden interactions.

9) Transformer-Based Models

The best fit for long-range dependencies in contextual relationship and large-scale financial data, while reducing false positives.

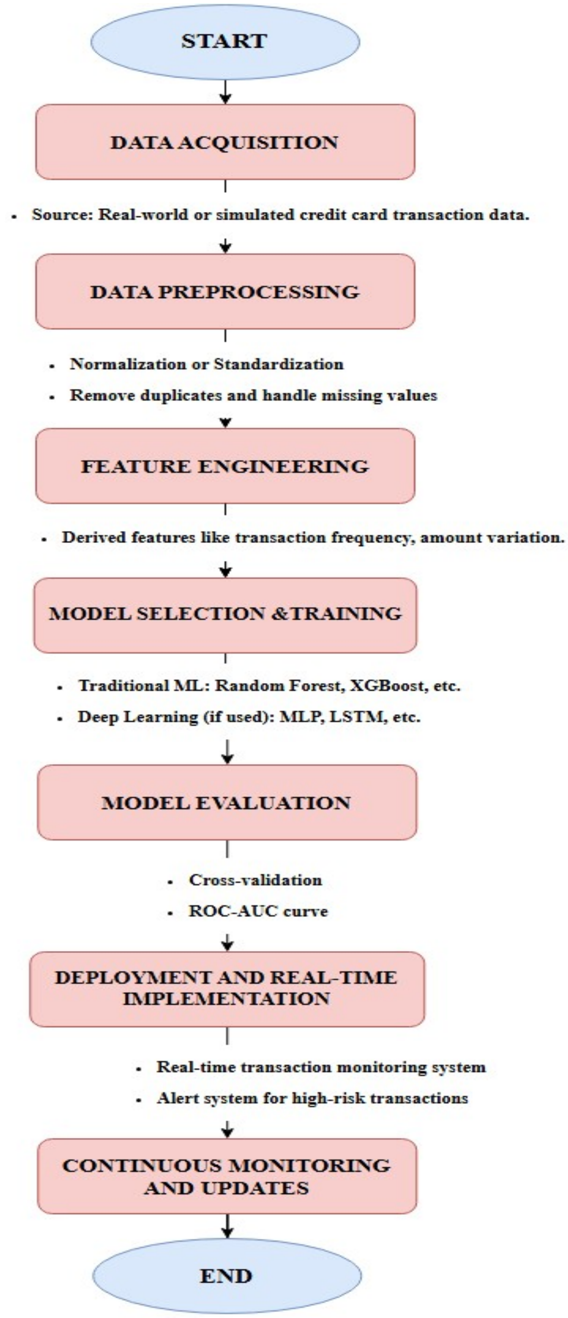

Fig. 2: Flowchart of Steps Followed for Implementation of AML Detection Model

D. Hyperparameter Tuning

Grid or random search techniques improve model parameters while improving predictive performance and generalization capability on financial claims data.

E. Cross-Validation and Evaluation Metrics

Cross-validation divides and partitions the data to create training and testing subsets multiple times to assess how well your model is performing [22] [23] [24].

1) Mean Squared Error

Measures the average squared difference between predicted values and actual values. Hence the larger the MSE, the lower predictive accuracy.

2) R^2 Score

Measures the variance explained by the model ,the closer to 1, the better the predictive ability of the model.

3) Mean Absolute Error (MAE)

Is the average magnitude of the prediction error (How far off, on average, predictions are from actual values), regardless of direction. MAE has less sensitivity to outliers than MSE.

4) precision or positive predictive value

Is the proportion of true positives among the predicted positive results. Fewer false positives indicate high precision.

5) Recall

Assesses the proportion of actual positives that were correctly identified. Is important if the cost of misidentifying a true positive is high.

6) Accuracy

Percentage of correctly predicted cases. Accuracy can be misleading if the classes are imbalanced.

7) F1 Score

Indicates the balance of precision and recall, is useful where there are imbalanced data sets, and when both false positives and false negatives are expensive.

8) ROC-AUC

Comparative test between versions of a model to examine its ability to discriminate between classes across various thresholds. Higher values indicate better discrimination between classes.

V. Results

This interpretive review has looked at developments in Anti-Money Laundering using Machine Learning and Deep Learning and other approaches published between 2018 and 2025.

A. Year-by-Year Progress on AML Research Using ML/DL

Research into AML rapidly accelerated past 2020, due to the larger societal trends of digital finance technology and AI-based aspects of compliance enhancing stakeholders' understanding of AML. Beginning in 2018, the number of publications rose steadily, with especially rapid growth in the last two years, peaking through 2023 – 2024. Weber et al (2018) [25] was the first to contribute an efficient GCN-based AML system, indicating a departure from rule-based detection mechanisms toward intelligent (i.e., model-based) detection mechanisms, while Jensen and Iosifidis (2022) [26] emphasized the under-explored domain of graphic data, and advocated AMD's moving forwards with unsupervised and semi-supervised models as well as ASICs for the production of synthetic data.

That year, Assumpção et al (2022) [27] presented DELATOR, an AML model, based on GNNs, using a dataset of over 5 million transactions and without the use of known 'true' labels.

With increasing complexities in transactions and Yu et al. (2024) [28] designing a hybrid CRNIM model using convolutional and recurrent layers that achieved 92.3% accuracy and fed into anomaly detection performance of approximately 15% higher than prior non-hybrid models, we could say that mobile banking was established as a favorable method for cross-border monitoring. Fan et al. (2025) [29] evaluated mobile transactions through a theoretical model employing transformers for account profiling. Their model was a hybrid of user behavior with transaction text mining understanding, reduced detection time by roughly 40% and achieved over 94% accuracy on synthetic data. Their findings lend to the position in which AML systems must adapt to speed and variety in mobile banking.

B. Qualitative Trend Analysis of AML Research (2018–2025)

The desktop review depicted in Figure 3 demonstrates the progress of AML research using ML and DL between 2018–2025. Rather than relying on case counts, Figure 3 provides a guide on the overall research over time.

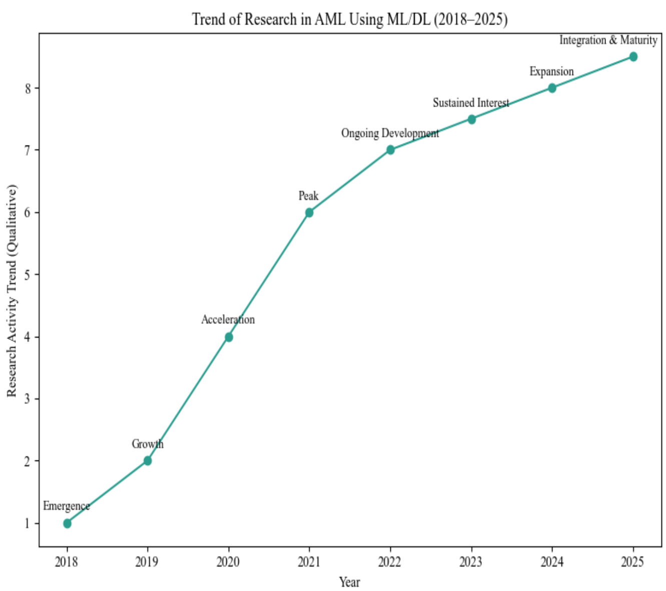

Fig. 3: Phases of AML Research Growth

The timeline of AI and ML in fintech begins with phase one, "Emergence" (2018), with earlier projects focusing on simple ML models (such as a decision tree). From there, in 2019, the second phase of the timeline, "Growth," continued to build upon the early-phase project, adapting and incorporating unsupervised learning and clustering of massive datasets [25]. From 2020, the third phase of the timeline, called "Acceleration", saw the introduction of temporal models and GANs, allowing for synthetic data creation [26]. Between 2021 and 2022, the timeline saw the introduction of "Peak" phase four with a wide adoption of GNNs and the introduction of systems (like DELATOR) that are showing their scalability, interoperability, and overall useful value regardless of data input [30].

In 2023, the fifth phase of the timeline, "Sustained Interest," focused on hybrid and transformer models, which came with an increased focus on interpretability (e.g., SHAP, LIME). In 2024, the sixth phase of the timeline, "Expansion," went to address mobile transactions and DeFi statelessness. By 2025, the field had entered "Integration & Maturity," as shown in phase seven of the timeline, with real-world deployment and transboundary AML models.

VI. CHALLENGES OF ML/DL IN AML

Regardless of the growing effectiveness of machine learning and Deep learning in anti-money laundering systems, they are still facing various significant challenges like:

A. class imbalance issues

Money laundering transactions are comparatively rare compared to the legitimate ones, resulting in highly imbalanced datasets. This makes it difficult for the model to learn meaningful patterns without overfitting to the legitimate transaction class. As a result, many models have good accuracy scores but poor recall for the actual laundering cases.

B. Lack of Explainability

Regulatory bodies demand transparency in decision-making related to money laundering. Still, deep learning models, especially neural networks, are complex and have non-linear structures and hence are considered black boxes. The inability to explain why a transaction was considered suspicious limits the model's adaptability in real-world banking environments where all the decisions are supposed to be auditable.

C. High False Positive Rates

Most AML systems focus on sensitivity over specificity to avoid ignoring any suspicious activity. Though this results in a much larger number of false positives that human analysts must investigate manually. High rates of false positives can reduce the system's efficiency and increase the operational cost.

D. Data shortage and privacy concerns

AML data is really sensitive, and access to the real-world data is not allowed because of the privacy laws, banking regulations, and customer confidentiality. Most of the banking bodies and institutions are restricted from sharing their customers' data, making it difficult to collect the data for training and testing the models.

E. Lack of standardized benchmarks

There is no universally accepted benchmark dataset and performance metrics for AML models, making it difficult to compare the results of different existing studies and papers.

VII. Future Directions

Future developments within anti-money laundering (AML) will advance on the back of privacy-preserving technologies, explainable processes, and real-time intelligent systems. Employing federated learning, multiple financial institutions can train various AI models in collaboration without sharing raw data, enhancing user privacy outcomes and helping establish a larger industry collaborative framework to fight financial crime and promote enhanced consumer trust. XAI generates the reasoning required for complex ML models, enabling institutions to demonstrate full compliance with regulatory requirements. Another achievement is enhanced transparency that improves the identification of false flags or threats classified as unidentified; however, GNNs offer tremendous potential for identifying suspicious transactions in payment networks using advanced pattern recognition methods. GNNs provide the capability to represent large volumes of complex, interrelated, and or agent-related data, enabling organisations to extract the surface patterns and atypical money laundering behaviours among jurisdictions and across agents. AML detection can then be performed using real-time aggregation (i.e. Edge AI) capabilities, allowing for timely responses to suspicious transaction activity in mobile and cross-border transactions. Additionally, processing local data incurs little to no latency and provides additional user privacy, while also allowing for immediate and therefore more effective tactical response by business to suspicious transaction activity.

VIII. Conclusion

Graph models revolutionise the AML movement as they provide a means for scalable and intelligent anomaly detection in the context of cross-border transactions (i.e. large-scale transaction environments) using graph-based analyses. Graph models provide the means to combine supervised Machine Learning analysis with unstructured and structured Data Analysis; Graph models transform how organisations profile risk through examining networks of relationships to identify hidden relationships between customers and suspicious patterns.

Nevertheless, there are significant challenges that need to be addressed: the limited interpretability of ML and DL solutions, the inability of these solutions to adapt to developing laundering trends, and the lack of access to quality data stemming from regulatory and privacy restrictions.

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